



Challenges and Opportunities of Drug Delivery for Treatment of Alzheimer's Disease

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Abstract

Alzheimer's disease (AD) is a neurodegenerative disorder characterized by progressive deterioration in cognitive functions. It represents a global health concern with increasing prevalence and devastating outcomes for the quality of life that could ultimately lead to death. AD is associated with deposition of β -amyloid ($A\beta$) plaques and intracellular buildup of tau proteins forming neurofibrillary tangles (NFTs), which are the main characteristics for AD brain tissues. Approved AD therapy is based mainly on symptomatic relief, and conventional medicaments often fail due to either low bioavailability, limited solubility, or failure to cross blood–brain barrier (BBB). The complexity in AD pathophysiology opens windows for many therapeutic options. So, lecanemab was recently approved by FDA as the first disease-modifying therapy. However, drug delivery to the brain remains challenging due to the nature of BBB. Hence, more extensive research is essential to develop disease-modifying therapies and also to find drug delivery strategies to ensure simplified administration and successful brain delivery. This review article summarizes AD pathogenesis with the corresponding treatment targets. It emphasizes innovative drug delivery strategies and novel formulation approaches to deliver medicines across BBB. The use of recent advancements in drug delivery to deliver medicaments across BBB are highlighted, with focus given to novel drug delivery systems and formulation of nanoparticles for brain targeting. The use of nutraceuticals, gene therapy, and stem cell therapy are covered.

Keywords Alzheimer's disease · brain targeting · blood brain barrier · intranasal delivery · nanoparticles

Introduction

Dementia is a group of illnesses distinguished by loss of cognitive capability leading to impairment in the ability to perform daily activities. Alzheimer's disease (AD) is considered to be the most common type of dementia, accounting for about 60 to 80% of dementia [1]. Alzheimer's disease is an irreversible neurodegenerative disorder that gradually impairs patient's memory and cognition. It particularly affects the geriatric population [2]. It is represented by gradual mental dysfunction, abnormal cognitive performance, personality and behavioral changes, speech impairment, and diminished verbal fluency [3].

Globally, by 2021, more than 57 million people suffer from dementia with higher prevalence in women [4, 5].

Alzheimer's Association estimated about 7.2 million Americans at the age of 65 and older could suffer from AD in 2025. The number is postulated to rise to about 13.8 million by 2060 unless medical advancements are made to retard or prevent disease progression. Rates of AD increase significantly with age, and incidence is suspected to rise in the near future [6]. Although uncommon, AD can affect populations under the age of 65 "early-onset" where the disease can progress differently. Early-onset AD markedly affects various cognitive domains, including motor ability and memory. Early-onset AD could be more aggressive than late-onset AD. Early-onset dementia (EOD) may develop due to various causes, including AD, frontotemporal dementia and vascular dementia. Other causes include Parkinson's disease [7]. EOD is assumed to have genetic basis, where gene mutations in presenilin 1 (*PSEN1*), presenilin 2 (*PSEN2*) and amyloid precursor protein (*APP*) increase production of longer and more aggregation-labile forms of $A\beta$. Mutations in *APP* gene can result in overproduction of $A\beta$ peptides, increasing the likelihood of $A\beta$ aggregation and amyloid plaques formation. Mutations in the *PSEN1* and *PEEN2* genes lead

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to abnormal APP processing with subsequent increased A β production. *PSEN1* variations could be responsible for about 30–50% of familial early-onset AD cases. Apolipoprotein E (*APOE*) gene has three alleles (ϵ 2, ϵ 3, and ϵ 4), from which APOE ϵ 4 allele represents the most important risk factor for AD. APOE ϵ 4 carriers also exhibit increased neuroinflammation as well as neuronal damage compared to noncarriers. Moreover, APOE ϵ 4 allele promotes tau pathology [8].

Alzheimer's disease (AD) was ranked to be among the leading causes for morbidity and mortality globally. By 2019, dementia was ranked as the 7th leading cause of both disability and death in the Middle East and North America. For individuals aged 70 and higher, dementia is categorized as the 3rd leading cause of disability and death [9].

Although AD etiology is not fully understood, many risk factors contribute in its progression. Risk factors for AD encompass genetic, environmental, behavioral, social, biological and clinical factors. Social and behavioral risk factors include physical inactivity, smoking and lower educational attainment. Clinical health conditions that may account for AD include obesity, diabetes, increased cholesterol and hypertension. However, combined behavioral, social, biological and clinical factors are together involved [10].

The pathophysiology of AD includes massive synaptic loss together with neuronal death in brain tissues responsible for cognitive functions. The existence of β -amyloid (A β) aggregates and neurofibrillary tangles (NFTs) produced by phosphorylated tau proteins are the main characteristics of AD. These aggregates are highly toxic, resulting in synaptic dysfunction due to inflammation as well as oxidative stress. Moreover, there are mitochondrial dysfunction, decreased levels of acetylcholine, and induction of glutathione depletion in brain tissues of AD patients. Each of these factors forms a foundation for specific treatment strategy and gives potential targets for medicaments [11].

Despite exploitation in research, successful treatment for AD is still elusive. At present, most approved drugs would provide only symptomatic treatment with no disease-modifying ability. Management of AD by these drugs involves the use of medicaments that target certain pathogenic pathway involved in AD progression. The treatment that would stop, hinder, or reverse AD progression represents an important health goal. Recently, a great focus was given to targeting beta-amyloid (A β) protein by monoclonal antibodies [1].

Approved medicaments for AD involve Acetylcholinesterase (AChE) inhibitors that work by increasing acetylcholine levels in brain favoring nerve cells communicating. Another mechanism is to inhibit N-methyl-D-aspartate (NMDA) receptors noncompetitively using memantine while the most recent approach is to target amyloid β aggregates using monoclonal antibodies. Despite the existence of such medicaments, AD management by such drugs remains challenging due to numerous factors. Failure of AD therapy

could be due to late treatment initiation, dose-dependent adverse effects and challenges associated with BBB permeability [12]. While BBB shields the central nervous system (CNS) from potential toxic pathogens and neurotoxins, it also hinders passage of many therapeutic drugs. Then, innovative therapeutic strategies are needed to overcome drug delivery challenges. Modulation of dosage form and administration route as well as the use of nanoparticles (NPs) to encapsulate therapeutic bioactives and target them into CNS can improve drug delivery for successful management of AD [13].

The current review discusses AD pathogenesis for possible selecting the most suitable disease targets for AD medicaments. Food and Drug Administration (FDA)-approved drugs, challenges with AD therapy and reasons for therapeutic failure are also addressed. Moreover, novel AD therapeutic strategies, the use of nutraceuticals, modified dosage forms and nanocarriers for AD therapy are explored in detail.

Pathophysiology of AD; The Way for Successful Treatment Strategy

Progression of AD involves a variety of complex mechanisms and understanding the molecular mechanisms for AD pathogenesis can help to find the most suitable molecular targets for AD management. Two main hypotheses were proposed for the pathophysiology of AD: the first one is amyloid cascade neurodegeneration, while the second is related to cholinergic system dysfunction. Other mechanisms include, but are not limited to, tau aggregation, oxidative stress, glutamate excitotoxicity, mitochondrial dysfunction, neuroinflammation and metal-mediated toxicity [11].

Amyloid Cascade Theory

According to this hypothesis, AD starts with amyloid precursor protein (APP) proteolytic cleavage leading to production, assembly, and deposition of β -amyloid (A β) plaques. In this hypothesis, the normal clearance process for A β is disrupted, leading to A β accumulation with A β plaque deposit formation. Accumulation of A β protein triggers a series of events in a cascade manner, eventually leading to death of nerve cells, damage of brain tissues, and finally causing AD symptoms [14].

Cholinergic Theory

In cholinergic hypothesis, dysfunction in cholinergic system can be sufficient to give rise to memory defects. In AD patients, there is cholinergic neuron degeneration together with a decline in acetylcholine level. Neuron degeneration

is attributed to synaptic loss, which in turn leads to memory impairment. Acetylcholine has significant roles in different cognitive functions as learning and memory. Increased AChE activity is a leading cause of reduced levels of ACh, a fact that supports the use of AChE inhibitors for AD therapy [15].

Tau Protein Theory

Tau is a key protein for nerve cell function, which stabilizes the microtubules in healthy neurons. Tau protein is phosphorylated abnormally in AD, leading to tau detachment from microtubules and formation of NFTs that lead to nerve cell death, a major factor in cognitive impairment [16].

Oxidative Stress

The brain is particularly susceptible to oxidative stress because of elevated oxygen consumption by brain tissues. Elevated reactive oxygen species (ROS) level impairs A β clearance and is also responsible for significant damage of important cellular components including DNA, resulting in disturbed homeostasis, cellular dysfunction and cell death [17].

Metal Ion Imbalance

Metal imbalance contributes to a variety of disorders, including neurodegenerative disorders. Redox transition metals, especially iron (Fe) and copper (Cu) can alter brain's normal physiology, and their levels are elevated in AD. Copper can promote A β aggregation, and its complexation with A β leads to neurotoxic hydrogen peroxide formation. Elevated copper levels also induce mitochondrial damage and impair cellular metabolism. This effect not only worsens oxidative stress but also enhances A β and tau protein aggregation. This theory introduces chelating agents as potential AD therapy [18].

Neuroinflammation

This hypothesis postulates that chronic inflammation triggers AD progression, where inflammatory mediators in brain tissues could lead to neuronal damage and impaired cognition. The CNS resident immune cells (microglia) have protective function in reducing A β plaques via phagocytosis of these aggregates. Inflammatory mediators lead to microglia overactivation, leading to its failure to resolve inflammation that in turn worsens neurodegeneration [16].

Targets of Alzheimer's Disease

Based on detailed understanding of AD pathophysiology, numerous targets for AD can be predicted. In brains of AD patients, there are extracellular amyloid plaques,

intracellular NFTs, abnormal tau protein phosphorylation, lowered acetylcholine, and mitochondrial dysfunction. In addition, there are oxidative stress and neuroinflammatory changes. Possible correlation between AD pathogenesis and possible treatment strategies and targets is presented in Fig. 1.

Acetylcholinesterase

AChE is the enzyme responsible for hydrolysis of acetylcholine. Its role is important for normal cell signaling, as well as for preventing neuron overstimulation. AChE lowers cholinergic activity, which is linked to cognitive impairment. Inhibition of this enzyme would restore normal levels for ACh. Hence, AChE inhibitors are thought to slow down AD progression. Examples of AChE inhibitors for AD include donepezil (DPZ), rivastigmine (RIV), as well as galantamine (GAL) [19].

NMDA Receptor

N-Methyl-d-aspartate (NMDA) receptor has a vital role in learning and memory. It allows passage of (Ca²⁺) across cell membrane for neurotransmission. However, excessive activity of NMDA receptor in AD causes excess entry of Ca²⁺, excitotoxicity, and cell death. Hence, it is important to find NMDA receptor antagonists that can inhibit excessive NMDA activation without interrupting NMDA's normal function. In this regard, memantine (MEM) is an approved non-competitive NMDA receptor antagonist commonly used to treat AD [20].

A β Plaque

A β plaques are considered to be the most important hallmarks in AD and their targeting can achieve promising therapeutic outcomes. Monoclonal antibodies (aducanumab, lecanemab) were recently produced to hinder A β plaque accumulation in brain tissues. Aducanumab successfully decreases levels of amyloid plaques in dose-dependent pattern, and patients receiving aducanumab showed improved cognitive functions [21].

Glycogen Synthase Kinase (GSK3)

GSK-3 β is involved in tau production with subsequent neurodegeneration. So, GSK-3 β inhibitors could hinder tau hyperphosphorylation. Tau targeting can also be considered for AD therapy by preventing the formation of NFTs, which are known for their significant role in neurodegeneration [22].

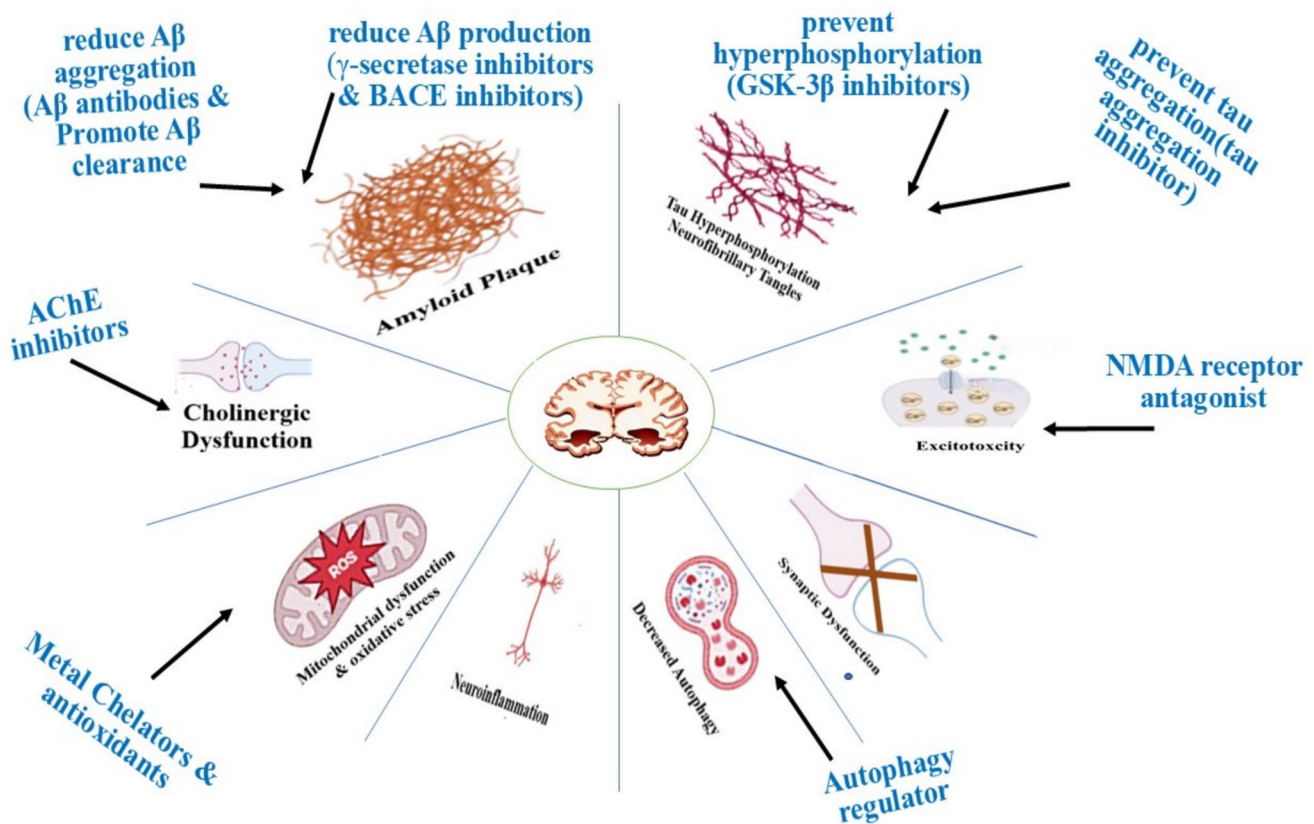


Fig. 1 Correlation between Alzheimer's Disease pathogenesis with corresponding treatment strategy

BACE1 (β -site Amyloid Cleaving Enzyme)

Abnormal processing of APP leads to accumulation of A β neurotoxic forms into plaques, which contribute to neuronal dysfunction and death. Three major secretase enzymes are involved in APP processing; α -secretase, β -secretase, and γ -secretase. From these enzymes, β -Secretase (BACE-1) is the protease which cleaves APP into A β peptide: therefore, inhibitors for this enzyme were investigated. However, such inhibitors showed high potency *in vitro* but have poor bio-availability and limited passage across BBB due to their high molecular weights [19].

While AD treatments involve medicaments that target more than one mechanism, certain targets may be identified for certain disease stages. AD has more than one stage, ranging from preclinical to severe AD. First, asymptomatic phase is characterized by absence of memory and cognitive loss. At the second stage (subjective cognitive decline), patients have subtle memory changes. The brain in both stages is characterized by early pathological alterations in cortex and hippocampus. Treatment that targets tau or amyloid could be effective at these stages since absence of symptoms indicates limited irreversible damage. Hence, AD treatment targeting tau or amyloid could be able to slow disease progression [23].

Patients with mild cognitive impairment (third stage) show symptoms of cognitive impairment such as loss of concentration or memory loss. The brain at this stage is characterized by A β accumulation, atrophy and tangle deposition in addition to neurodegeneration. FDA approved anti-amyloid would be prescribed at this stage [24]. In the next stage (mild dementia due to AD), patients have mild impairment in their daily activities. At this stage, brain loses about 15–25% of synapses [25]. Treatments recommended include cholinesterase inhibitors and anti-amyloid therapy. At the moderate dementia stage (fifth stage), disease spreads to cerebral cortex areas resulting in increased memory loss. Amyloid distribution in the brain is highly pronounced, tau accumulation increases to its maximum and medial atrophy and neurodegeneration become more evident. In this stage, patients need help with their daily activities. Treatment includes the use of combination therapy (memantine and cholinesterase inhibitors) [26]. In severe dementia stage, disease spreads to the entire cortex with severe accumulation of plaques and NFTs resulting in progressive cognitive impairment where patients aren't able to perform daily activities. For this stage, cholinesterase inhibitors together with memantine are prescribed. Moreover, focus is given to management of other symptoms including sleep disturbance, agitation and psychosis [27, 28].

Pharmacological Approaches for AD Therapy

Diverse pathological pathways involved in AD recommend the use of more than one treatment strategy to interfere with disease progression.

FDA-Approved Medicaments

According to AD pathophysiology, various targets for AD were predicted. However, two main classes of therapeutics are approved for AD therapy, which are acetylcholinesterase (AChE) inhibitors as well as NMDA receptor antagonists. Table 1 presents information regarding these classes together with the recently approved medicaments, which involve monoclonal antibodies.

Acetylcholinesterase (AChE) Inhibitors

Based on cholinergic hypothesis, the root cause of AD was assumed to be the decreased ACh production. Hence, AChE inhibitors could be a promising approach for improved

cognitive functions. AChE inhibitors involve oral formulations of tacrine, rivastigmine, galantamine, donepezil and also rivastigmine patch. These medicaments work by inhibiting AChE to compensate ACh levels for postsynaptic stimulation. AChE inhibitors have some gastrointestinal side effects (nausea and vomiting), which may result in discontinuation of treatment.

NMDA Receptor Antagonist

NMDA receptor was identified as a potential target for the treatment of AD based on the assumption that its excessive activity results in neuronal excitotoxicity. Memantine is NMDA antagonist with low affinity, so it can be displaced from receptors by high glutamate levels. Thus, it inhibits excitatory receptors without either induction of persistent blockage or hindering normal synaptic transmission [35].

Other Approved Medicaments

Monoclonal antibodies can be used to target A β fibrils in patients with early AD. In this concern, monoclonal antibodies (lecanemab or aducanumab) were recently approved as

Table 1 Approved Therapeutics for Alzheimers Disease

Drug	Drug description	Approved use	Side effects	Key information
Acetylcholine esterase inhibitors				
Tacrine [29]	First approved AChE inhibitor	Mild to moderate AD	Hepatotoxicity limited its use to certain cases	Short half-life necessitating frequent administration
Donepezil [30]	Reversible piperidine-based AChE inhibitor	Mild to moderate AD	GIT problems and sleep disturbance	-Transdermal patches are available for once week therapy -Half-life: about 50–70 h
Rivastigmine [31]	Pseudo-irreversible antagonist	Mild to moderate AD	GIT problems after oral use	first pass effect -Half-life of oral dose is 1–2 h, and 3–4 h for transdermal
Galantamine [31]	Alkaloid from plant source	Mild to severe AD	GIT problems, headache and dizziness	Available as immediate release tablets or oral solution with half-life (6–8 h)
NMDA receptor antagonist				
Memantine [32]	Regulate glutamate Neurotransmission, prevent excitotoxicity	Moderate to severe AD	Dizziness, confusion, constipation and vomiting	Its plasma half-life is about 60–100 h
Combined therapy				
Memantine + Donepezil [33]	AChE inhibitor + NMDA receptor antagonist	Moderate-to-severe AD	Constipation, nausea, increased bowel movements, headache, disorientation	This treatment provides simple treatment regimen and reduces pill burden
Recently approved				
Aducanumab [11]	Selectively targets β amyloid aggregates	Moderate AD	Headache and diarrhea and Infusion-related reactions	It induces amyloid-related imaging abnormalities as cerebral edema
Lecanemab [34]		Mild AD		

drugs with the ability to slow down cognitive decline after intravenous infusion every two weeks. These drugs bind selectively to A β forming a complex that is then cleared by phagocytosis. However, infusion-site reaction and amyloid-related abnormalities are associated with its use and observation of long-term side effects is required [36].

Sodium oligomannate (oligosaccharide from marine origin) was developed and approved in China for mild-to-moderate AD. The drug inhibits A β accumulation and thus inhibits neurotoxicity induced by A β aggregation. However, ongoing trials are done to further investigate its efficacy and safety [37].

Non-Approved Medicaments

The complexity of AD progression hampers therapeutic efficiency, resulting in increased global burden for AD. Currently, different drugs are being investigated to target AD pathways. However, such bioactive substances are still under varying investigation phases. Examples include anti-amyloid

therapy, BACE inhibitors, γ -secretase inhibitors and others. Moreover, many natural products are being explored for their benefits for AD by antioxidant, anti-inflammatory and neuroprotective characteristics [38]. Some examples for different molecules and monoclonal antibodies in clinical trials are listed in Table II.

Nutraceuticals and Plant-Based Ingredients

Nutraceuticals are food-derived components with medicinal benefits. Several substances of dietary origin have protective efficiency against some disorders, including neurodegenerative ones. Galantamine, a plant-derived alkaloid, was the first FDA-licensed nutraceutical for reversible AChE inhibition. Recently, more focus is given to developing food-based therapeutics to minimize AD progression and alleviate symptoms. A combination of modern bioactive components together with traditional food may promote the development of suitable disease-specific treatment regimens that hinder and cure several chronic diseases. Popular effective

Table II List of Novel Therapeutics in Different Phases of Clinical Trials for AD

Phase	Agent	Target	Mechanism	Trial no
Phase I	MK-2214	Tau	Target the phosphorylated tau to prevent tau spread	NCT05466422
	IONIS MAPTRX	Tau	Directly reduces neuronal tau levels	NCT03186989
	LY3202626	β Amyloid	BACE 1 inhibitor	NCT02323334
	MEDI1814	β Amyloid	Antibody reduces free A β_{42}	NCT02036645
	EVP-6124	ACh	Activates acetylcholine receptor	NCT00766363
Phase II	T-817MA	β Amyloid	Inhibit neuronal cell death induced by A β_{42} and H $2O_2$	NCT02079909
	ABBV-8e12	Tau	Decrease the spread of tau pathology in brains	NCT02880956
	Rilapladib	Inflammation	Lipoprotein-associated phospholipase A2 inhibitor	NCT01428453
	Varenicline	ACh	Stimulate nicotinic acetylcholine receptor	NCT00744978
	Bryostatin 1	β Amyloid	Activate α -secretase, thus preventing A β aggregation	NCT02431468
	VX-745	Synaptic dysfunction	Protein kinase inhibitor	NCT02423200
	Crenezumab	β Amyloid	Promote amyloid plaque clearance	NCT01723826
	CT1812	β Amyloid	Prevent A β oligomer binding	NCT04735536
	Troriluzole	Glutamate	Reducing synaptic levels of glutamate	NCT03605667
	PTI-125	β Amyloid	Restore the normal shape and function of neuronal filaments	NCT03748706
	AMX0035	Mitochondrial dysfunction	Reduce neuronal death and improve cellular function	NCT03533257
	BI 409306	Glutamate	BACE1 inhibitor	NCT02337907
	ELND005	β Amyloid	Inhibit β -amyloid aggregation	NCT00568776
	Ladostigil hemitartrate	ACh	Inhibit brain acetyl and butyrylcholinesterase	NCT01354691
	CX516	Synaptic dysfunction	Enhance synaptic responses	NCT00040443
Phase III	RVT-101	More than one target	Inhibit 5-hydroxytryptamine 6 (5-HT $_6$) serotonin receptor	NCT02585934
	Semagacestat	β - amyloid	γ -secretase inhibitor, prevent A β production	NCT01035138
	Simufilam	β - amyloid	Restore shape and function of Filamin A protein that forms clumps in AD	NCT04994483
	Donanemab	β - amyloid	Promote A β plaque clearance	NCT03367403
	TRx0237	Tau	Tau aggregation inhibitor	NCT03446001
	Solanezumab	β -amyloid	Promote A β plaque clearance	NCT02008357
	Suvorexant	β -amyloid	Reduce A β deposition in the hippocampus and cortex	NCT02750306

nutraceuticals could have antioxidant, autophagy regulation and anti-inflammatory properties in addition to other beneficial roles. Examples include garlic extract, turmeric and other plants. However, it is essential to perform detailed clinical trials [39, 40].

Curcumin (*Curcuma longa* product) has typical anti-inflammatory, anti-amyloidogenic and anti-tau functions. Therefore, curcumin can be used in neurological illnesses after incorporation into suitable drug delivery system to enhance its pharmacokinetics and pharmacodynamics [41]. Piperine has also exhibited improved memory impairment in rats with AD [42]. Another example includes quercetin, which was shown to improve memory defects and cognitive functions in AD [43].

Metal Chelators

The evidence for the role of oxidative stress and metal dys-homeostasis in AD pathology introduces metal chelators as an approach for AD management. Metal chelators can capture metal ions to interfere with their participation in A β aggregation induced by redox processes. Many coumarin derivatives can chelate metal ions while having antioxidant and anti-inflammatory properties. Other examples of metal chelators include clioquinol, deferiprone and tenuazonic acid. The ability of metal chelator to cross BBB can be enhanced by the use of nanoparticles [44].

Other Medicaments

Estradiol (17 β) was found to have neuroprotective functions, suggesting its possible use for dementia. It was also found to reduce neuroinflammation, decrease brain oxidative damage and increase cholinergic functions. However, a suitable formulation is needed to overcome its poor solubility [45]. Nicotine was also found to have neuroprotective properties, where it provides protection against neurotoxicity caused by glutamate and A β [46].

Challenges for AD Management

Challenges for the management of AD include both challenges for effective therapeutics as well as challenges for treatment persistence and patient compliance. Therapeutic treatment of AD remains complex due to diverse factors involved in its pathology. Many factors are involved in its pathogenesis, including inflammation, cerebrovascular disease and traumatic brain injury. Based on this complexity, many difficulties are encountered in developing successful treatment options [47].

Therapeutic treatment was based on the use of medicaments such as AChE inhibitors and NMDA antagonists, as explained before. Clinically, the efficacy of most approved

drugs in enhancing cognitive function is poor and such medicaments are not able to stop disease progression. AD patients are highly susceptible to missing doses due to the nature of the disease. That is why development of successful dosage regimens that deliver drugs to the target site, enhance patient compliance and avoid multiple daily doses is necessary [48].

Most approved medicaments are available as oral dosage forms that have several limitations, such as lacking brain targeting and having first-pass effect with corresponding poor bioavailability. Therapeutic failure can develop due to unfavorable pharmacokinetics as well as pharmacodynamics of medicaments. Other drug-related factors include unfavorable drug solubility (hydrophobicity), high molecular weight, insufficient absorption via biological membranes and poor drug stability or tissue toxicity (neurotoxicity, hepatotoxicity, or kidney toxicity) [49, 50].

AD is a neurological illness affecting the brain, which is the most protected organ. Brain is surrounded by multiple protective barriers that increase the difficulty for targeting therapeutic agents to the brain. BBB prevents passage of 98% of pharmaceuticals to the brain, which necessitates the development of effective platforms for AD treatment. Although there is great development in nanomedicine, BBB can hinder their passage to the brain. This explains the need to decorate nanomedicines to enhance their brain accumulation [51]. Multidrug-resistant proteins such as P-glycoprotein (P-gp) could lead to therapeutic failure by preventing medicaments from achieving their therapeutic goals through reducing their distribution in brain tissues [52].

Although the use of anti-amyloid monoclonal antibodies provides a promising chance for AD therapy due to their ability to hinder amyloid plaque formation, their clinical efficacy remains controversial. Several challenges have to be addressed for their use in clinical practice, such as route of administration, the need for imaging as well as genetic testing and their potential to cause serious adverse events [48].

To overcome these limitations, several strategies were proposed in recent studies, including modification of dosage forms and route of administration, the use of nanomedicine, repurposing some medicaments, the use of gene therapy and the possible effect of immunization.

Novel Strategies to Overcome Challenges of AD Delivery

Numerous strategies were utilized to effectively deliver AD therapy to the target tissues. Invasive techniques, such as disrupting BBB integrity lack patient acceptability while

harming the neuroprotective performance of BBB. Disrupting BBB integrity aims to disrupt brain endothelial cell integrity to facilitate entry of various molecules into cerebral tissue. Disrupting BBB integrity can be achieved by electroporation using pulsed electrical field to disrupt electrical potential across cell membrane. Other invasive techniques of reduced patient acceptability include infusion by intrathecal or intracerebroventricular administration. Non-invasive techniques include the use of dosage forms to achieve nose-to-brain delivery. Other approaches include inhibiting efflux transporters and the use of nanocarriers (Fig. 2). From these strategies, the use of nanocarriers can be considered the most suitable with promising outcomes [53].

Modulation of Dosage Form Design and Route of Administration

For AD, extended-release (ER) formulation provides the advantage of reduced number of doses. Transdermal patches as well as microneedles could enhance patient compliance by maintaining extended release while

improving drug delivery by escaping first-pass metabolism. Nasal administration is also investigated for direct brain delivery. Figure 3 illustrates examples for dosage forms investigated for AD.

Oral Extended Release (ER) Formulations

ER formulations involve delivery systems controlling drug release by diffusion, erosion or osmotic mechanisms using osmotically active ingredients. Despite the risk of reduced bioavailability and dose dumping, oral ER formulations have advantages of reduced dosing frequency, smooth drug concentration, uniform drug effect by stable plasma levels and increased compliance [54]. Hence, memantine ER once-weekly capsules were developed as more convenient dosage that would improve patient adherence to therapy. Sustained MEM plasma levels over seven days were achieved by varying formulation ingredients (polycaprolactone, Eudragit, ethyl cellulose). Pharmacokinetic studies confirmed smooth plasma MEM levels over 7 days with higher T_{max} than marketed daily formulation [55].

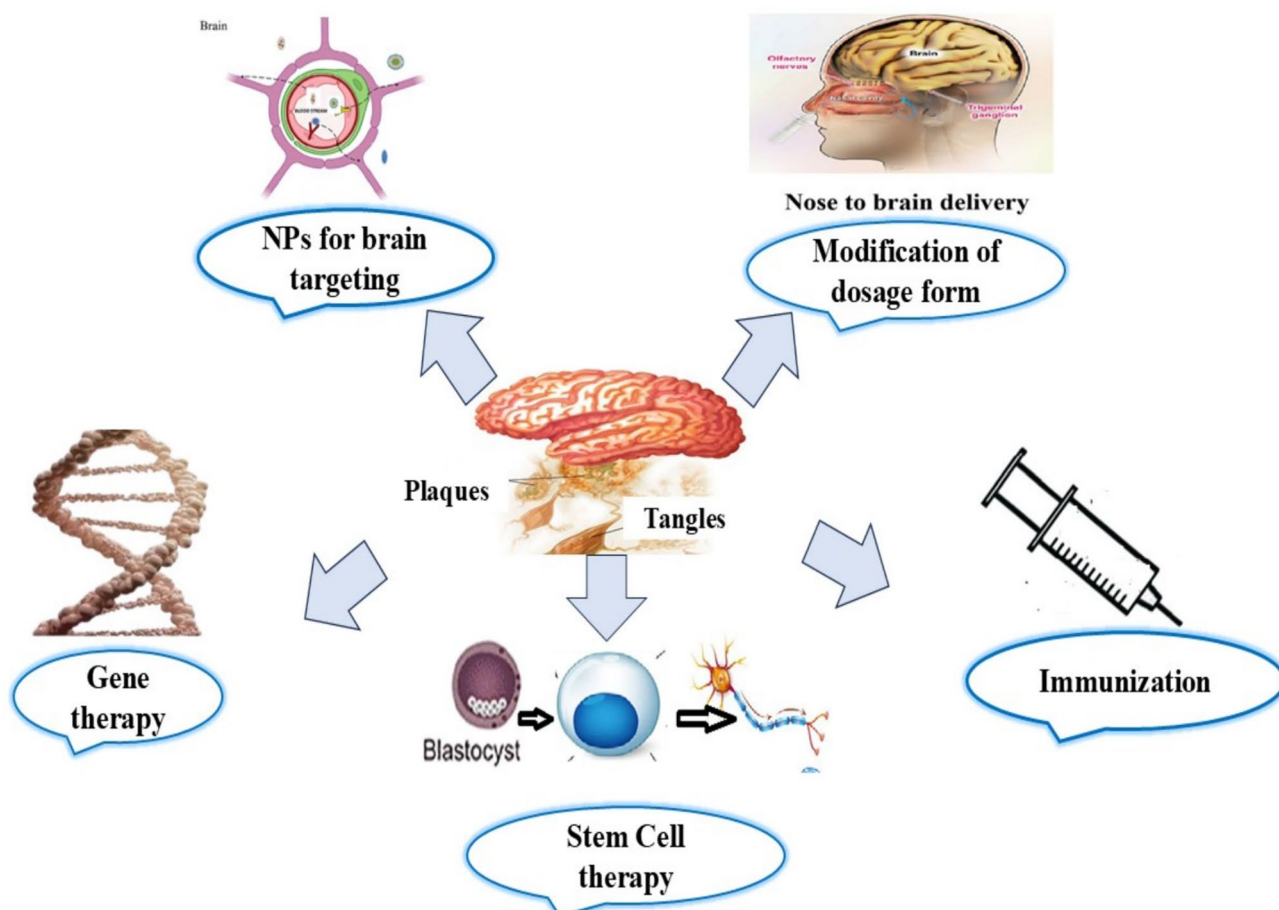
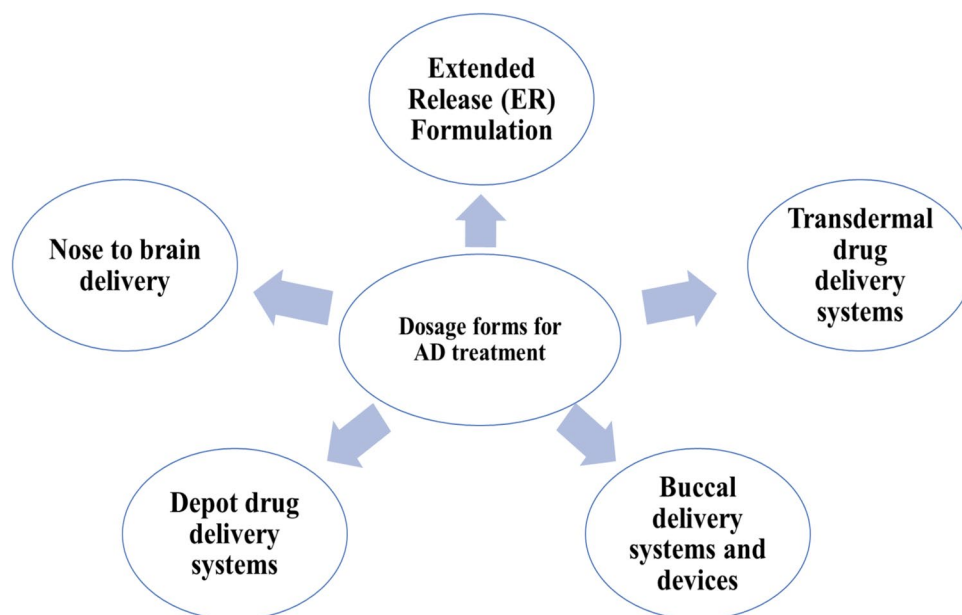


Fig. 2 Novel approaches for Alzheimer's disease therapy

Fig. 3 Different drug delivery systems for AD therapy



Buccal Drug Delivery

Patients at moderate or severe AD stages have swallowing difficulty. So, development of orally disintegrating tablets (ODTs), oral disintegrating films (ODFs), or mouth dissolving films (MDFs) could be a better option since these dosage forms could rapidly disintegrate after being placed on the tongue. These formulations are particularly useful if first-pass effect is to be avoided. In cases when drug is to be available for extended absorption in GIT, coated pellets can be first prepared and used for ODT formulation. Despite being more expensive than tablet/capsule formulations, several advantages could be provided by these formulations such as being taken with no water or chewing needs and improved taste, provided that the drug is not bitter [56, 57].

ODT formulations for donepezil were approved to favor administration. Unfortunately, the bitter taste of these ODTs required taste-masking investigation for enhanced patient compliance. In a study by Sutthapitaksakul and coworkers, taste-masked DZP microparticles were developed by double emulsion solvent evaporation method to be used for ODTs preparation. The developed ODTs had acceptable *in vitro* dissolution and improved palatability [58]. In another study, DZP ODTs were prepared by simple wet-granulation method where three different grades of lactose monohydrate were used to mask its bitter taste. Results showed promising release characteristics and palatability [59]. Also, ODFs developed by solvent-casting technique were studied for rivastigmine delivery and pharmacokinetic results showed improved drug bioavailability compared to RIV reference capsule [60].

Transbuccal Delivery

Transbuccal delivery uses chemico-physical mechatronic approaches to release drugs through transbuccal route. This approach can theoretically overcome limitations encountered by conventional oral formulations. Users can control the device to monitor drug level. However, the main drawback is the need for two missing adjacent teeth in the patient's mouth. In this concern, transbuccal device (IntelliDrug) for galantamine administration was proposed. The device is to be implanted in oral cavity by specific techniques. The device has drug reservoir, drug level sensor, power source and a mechanism for pushing drug solution. Results suggested that this delivery device can generate controlled blood levels for galantamine [61].

Transdermal Delivery

Transdermal delivery offers an acceptable option for patients with swallowing difficulty or non-adherence to therapy. This drug delivery can promote patient compliance by allowing regular intake of drug in patient-friendly manner. This route also avoids first-pass effects, provides smooth plasma level and allows reducing daily doses. However, this delivery system can be quite expensive, irritant or sensitizing to skin and not suitable for drugs requiring high blood levels. Transdermal patches are from these delivery regimens, which have additional advantage of possible cessation of drug delivery upon appearance of side effects. However, patch removal is difficult from certain body areas such as the back. It is also recommended to change application site daily without using the same skin area [62].

Rivastigmine (Exelon®) patch, a matrix diffusion-controlled patch, was developed and approved for AD treatment. After a lag time (0.5–1 h), RIV blood level in optimal therapeutic window was obtained. The patch is well tolerated by patients with AD with gastrointestinal effects much lower than that for RIV oral formulations. In a study by Schurad B. *et al.*, a twice weekly RIV patch formulation was developed to enhance patient compliance and its bioavailability was compared to once daily patch (Exelon®). Results confirmed bioequivalence between the formulations with favored patch adhesion for the new formulation [63]. From the caregiver point of view, transdermal RIV is more promising than oral medication in terms of enhanced patient satisfaction and improved adherence [64].

Ointment is another form of transdermal delivery. Phen-serine (AChE inhibitor) ointment was developed to tackle challenges associated with its first-pass effect. Formulation was examined, and selected transdermal ointment was evaluated *in vivo*. Results confirmed enhanced cognitive function in AD animal model. However, more investigations are needed for translating these findings into clinical use [65].

Microneedle patches (MNs) are from the recent transdermal delivery techniques where skin permeability is enhanced by forming micropores in stratum corneum with the help of microneedle arrays. For the delivery of donepezil, a novel dissolving MN patch was developed. Donepezil released after 1-h and 1-week applications achieved sustained DZP level compared to oral dose. Donepezil was also successfully detected in the plasma, illustrating the ability of MNs to enhance drug permeation making them promising candidates for DZP delivery to elderly AD patients [66].

In another study, coated MNs were developed for donepezil delivery. Physicochemical investigations confirmed successful DZP incorporation and suitability of the DZP-coated MNs for transdermal delivery. Permeation studies illustrated successful DZP permeation, while laser scanning microscopy confirmed DZP skin distribution and sustained DZP release [67].

Depot Drug Delivery

Injectable depot-forming preparations for subcutaneous or intramuscular injection are designed to release drug in controlled manner which can be extended up to 1 month. *In situ* forming implants are of great interest because of ease of manufacturing as well as ease of administration. They are prepared to be in sol form prior to administration but become macroscopic gels immediately after injection to provide sustained drug release. *In situ* gelation adopts the use of temperature modulation and pH change. Depot dosage forms avoid flocculated plasma drug levels, avoid first-pass effect and enhance patient acceptability by reducing dosing frequency. However, pain at injection site, administration

difficulties and inability to remove medicament once administered are considered the main disadvantages [68].

For the aforementioned advantages, subcutaneous injection of RIV was developed and evaluated by Lipp and coworkers for safety and efficacy. Results demonstrated prolonged RIV release for 7 days with AChE inhibition for 14 days. The results provide subcutaneous RIV injection as candidate for increasing dosing interval for AD patients [69].

The promising effect of *in situ* galantamine gel for intramuscular injection was confirmed in a recent study. *In situ* gel developed by Poly-dl-lactide-coglycolide, and was injected in rats for biodegradability study, while rabbits received the same formulation for pharmacokinetic study. Developed GAL *in situ* gel had acceptable biodegradability rate, while pharmacokinetic evaluation showed extended GAL residence up to 71 days [70].

Intranasal (IN) Delivery

Finding solutions to overcome limited drug transport across BBB is a central goal to prevent therapeutic AD failure. Recently, intranasal (IN) delivery systems were proposed as an alternative approach to systemic administration and brain targeting. IN represents non-invasive therapeutic strategy for CNS disorders, including AD. Nose-to-brain (N2B) was utilized for drug delivery across BBB due to unique connections between olfactory region and the brain. This route provides rapid drug action while avoiding first-pass effects. However, bypassing clearance mechanisms by metabolic enzymes and mucociliary systems and escaping efflux transporters in olfactory region are the main challenges for N2B delivery [49, 54]. Moreover, formulation viscosity also has to be considered. In an attempt to overcome administration difficulty of highly viscous gels while preserving enhanced retention, *in situ* gel systems using polymers with sol–gel behavior were explored [71, 72].

To achieve Rivastigmine IN delivery, nasal RIV spray was studied for promising RIV brain delivery. In this study, RIV brain delivery was studied via pharmacodynamic investigation. Nasal spray was found to be superior to commercial oral formulation, highlighting the success of IN delivery for RIV delivery [73].

In another study, tacrine *in situ* gel for IN delivery was developed using thermosensitive polymer (Pluronic F-127). Several factors were studied to optimize physical properties and nasal residence time. Pharmacokinetics as well as brain dispositions of the developed *in situ* gel were studied and compared to oral tacrine solution in rats. Results confirmed significantly prolonged drug retention in the nasal cavity as indicated by mucociliary transport time. In addition, pharmacokinetic study showed elevated drug bioavailability compared to oral solution. The enhanced brain uptake of tacrine introduces this formulation as an effective IN tacrine delivery system [74].

IN delivery could also be achieved using nasal films. In an interesting study, innovative donepezil films were developed for AD. Film former (Hydroxypropyl-methyl-cellulose), plasticizer (Polyethylene glycol 400) and permeability enhancer (Methyl- β -Cyclodextrin) were varied to achieve accepted film flexibility and DZP permeation across rabbit nasal mucosa [75].

Despite the successfulness of IN delivery, drug physicochemical properties provide obstacles to effective N2B delivery. Drugs of high molecular weight face difficulty to effectively pass through nasal mucosa. Furthermore, drug chemical stability in nasal environment must be taken into consideration. Some drugs may be degraded upon contact with nasal enzymes or nasal environmental pH. Also, volumes suitable for IN administration may restrict drug quantity that could be efficiently administered via IN route. To tackle these challenges, many techniques are adopted, such as the use of nanocarriers to optimize drug delivery for effective brain targeting [76].

Nanotechnology-Based Drug Delivery Systems

Therapeutic AD failure can develop due to unfavorable drug pharmacokinetics and/or pharmacodynamics. Drug development in the form of nanoplatforms can enhance its pharmacokinetics as well as pharmacodynamics. Owing to their reduced dimensions and increased surface area, nanoparticles (NPs) can permeate BBB via tight junctions through openings in brain capillaries' endothelial cells. NPs, especially surface-modified ones, have the ability to be transported via multiple mechanisms, including receptor-mediated transcytosis as well as adsorptive-mediated transcytosis. Mechanisms of transport across BBB are illustrated in Fig. 4.

An important benefit obtained in nanomedicine is controlling and targeting drug release into certain sites. Nanotechnology-based drug delivery systems include organic and metallic nanoparticles. Each of these can be developed as a promising tool for effective CNS targeting via a variety of routes, especially intranasal route. For drug delivery across BBB, (NPs) were adopted to increase drug permeability, absorption and retention in brain capillaries. NPs can also be decorated to escape phagocytosis and efflux system. Moreover, NPs can be tailored to target A β in cells. NPs can also be coated with tween 80, mucoadhesive polymers and certain

antibodies [77, 78]. Different types of NPs studied for AD therapy are illustrated in Fig. 5.

Similar to NPs, microformulations, including microspheres and microemulsions, were also studied for AD to improve drug efficacy and patient adherence. For example, donepezil-loaded poly-lactic-co-glycolic acid and polylactic acid microspheres were developed and loaded in injectable hydrogel system as promising approach to reduce dosing frequency [79].

Microemulsion-based lyotropic liquid crystalline (LLC) is an emulsion formed via self-assembly of mesophasic lipids. LLC provides a suitable approach for improved mucoadhesion based on phase transition from low-viscous microemulsion to more viscous systems. Donepezil administration via IN route was studied using mucoadhesive LLC. A colloidal formulation was developed, and pharmacokinetics were evaluated in rats. Significant DZP level was achieved in rat brain after IN administration of the developed microemulsion [80].

Despite the aforementioned benefits, there is a gap regarding biocompatibility of NPs. NPs' interaction with tissue carriers can trigger allergic reactions, and some NPs (silver NPs) could harm brain cells and disturb BBB integrity. So, comprehensive NP evaluation is required for

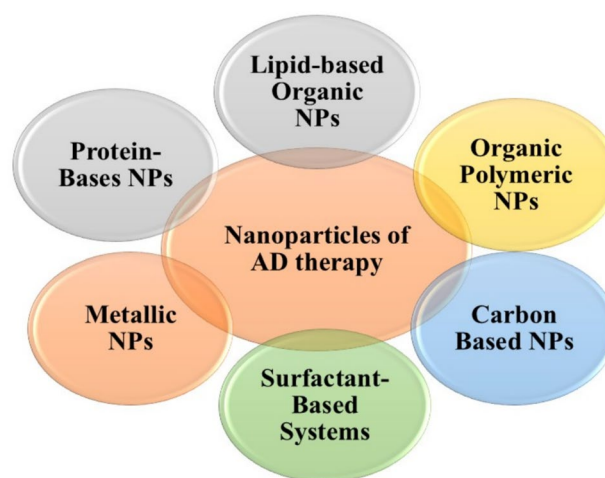
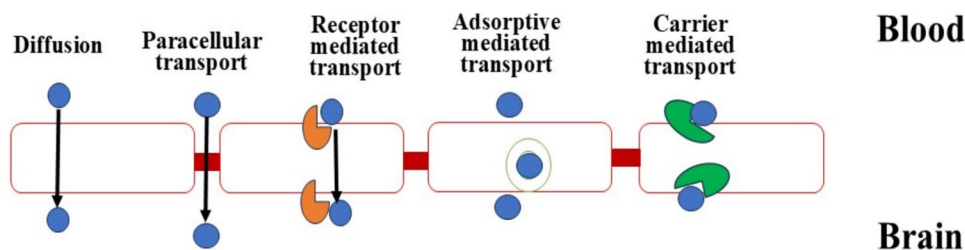


Fig. 5 Different types of nanoparticles studied for Alzheimer's disease therapy

Fig. 4 Schematic illustration of different transport mechanisms across BBB



accurate utilization of NPs [81]. This explains the absence of approved NPs for AD management.

Lipid-based Organic Nanoparticles

Lipid-based organic NPs for therapeutic purposes include liposomes, solid-lipid NPs (SLNs), nanostructured lipid carriers (NLCs), niosomes and others. SLNs have solid lipid core (glycerides, fatty acids or waxes) that is able to solubilize lipophilic drugs. The lipid core is stabilized by surfactants to prevent particle agglomeration. Memantine and tramiprosate SLNs were prepared by Shivananjegowda and colleagues to promote A β clearance in AD rat model after intraperitoneal injection. The core was made up of labrasol together with gelucire, while tween 80 as well as labrafil were adopted as surfactant and co-surfactant, respectively. SLNs were optimized to ensure drug encapsulation and adequate size. *In vitro* A β fibrillation together with molecular docking studies were used to confirm binding to A β . *In vitro* cell viability confirmed highly enhanced neuroprotection for NPs compared to pure drugs. Formulated SLNs had improved pharmacokinetic profile and higher brain accumulation than pure drug. Moreover, behavioral studies assured enhanced spatial memory, while histological investigations confirmed diminished A β plaques in rat hippocampus [82]. Also, piperine SLNs prepared by emulsification-solvent diffusion were coated with polysorbate-80 to enhance brain uptake. Results showed improved cognition in AD animal models after intraperitoneal administration as compared to DZP. Brain histopathology investigations confirmed reduced plaques as well as tangles [83].

Nanostructured lipid carriers (NLCs) are lipid NPs formulated with a core of a mixture of liquid lipid together with solid lipid. NLCs were developed by high-pressure homogenization together with ultrasonication for effective donepezil brain delivery via IN route in a recent study. Glyceryl monostearate was adopted as solid lipid, while argan oil was used as liquid lipid. Tween 80 and Poloxamer 407 were used as surfactant and co-surfactant, respectively. NLCs displayed good drug encapsulation with adequate size and zeta potential. Selected formulation achieved higher DZP brain concentration (9 times) after IN than DZP IV administration suggesting that DZP NLCs can be efficient drug delivery for DZP administration [84].

Liposomes are vesicular biocompatible NPs formed of phospholipid bilayers surrounding aqueous core and are able to deliver various medicaments across BBB. Liposomes can be additionally modified for brain targeting and accumulation by several methods, such as the use of transferrin (Tf), which enhances drug brain accumulation via binding to transferrin receptor (TfR) on BBB. In a recent study, transferrin-modified vitamin B12 liposomes were developed to enhance vitamin B12 bioavailability and benefit from its

anti-amyloidogenic effect in AD. Produced liposomes had suitable sizes (<200 nm), vitamin B12 encapsulation, and zeta potential suitable for brain delivery. In addition to their ability to disrupt mature fibrils, the developed liposomes were able to delay A β fibril formation (as indicated by thioflavin-t fluorescence assay), highlighting their potential for prevention as well as AD treatment [85].

Rivastigmine liposomes, as well as modified liposomes (using cell-penetrating peptide, CPP), were developed by Yang *et al.* to improve RIV brain distribution after IN administration. Results exhibited that developed liposomes, especially modified ones, enhance RIV permeability across the BBB *in vitro*. IN administration of these liposomes enhanced RIV brain distribution and retention compared to intravascular RIV. The study suggested that these liposomes can be used to improve brain delivery after IN administration and at the same time decrease hepatic first-pass effect and GIT adverse events [86].

Niosomes are also vesicular NPs composed of self-assembled nonionic surfactants in aqueous environment. Compared to liposomes, niosomes can reduce drug flux and hence are considered to be more stable than liposomes. In a recent study, synthesized MEM derivatives-encapsulated niosomes were used for enhanced BBB permeability in order to minimize the drug's peripheral toxicity. Results showed effective drug encapsulation, adequate stability with controlled drug release. *In vitro* BBB permeability study was used to confirm the formulation efficacy. Niosomal formulations promoted BBB permeability compared to free drug derivatives. This study's results provided a promising platform to overcome limitations by BBB for the success of AD treatment [87]. In another study, niosomal NPs of rivastigmine were prepared for better brain targeting. Film hydration was used to successfully encapsulate RIV in niosomal NPs using different grades of Span and Tween. Results proposed that these vesicles can be promising candidates for RIV delivery after comprehensive investigation in animal models [88].

Transferosomes are lipid-based vesicles proposed especially for transdermal administration. They are highly flexible, deformable vesicles formed mainly of phospholipids in addition to edge activators. Edge activator enhances transferosome permeability across stratum corneum [89]. Transferosomes were studied for encapsulating donepezil in a recent study where hyaluronic acid-coated DZP transferosomes were prepared for N2B for brain to overcome its oral GIT adverse effects. Film hydration was used in the preparation, and formulation was optimized for suitable diameter, drug encapsulation and *ex vivo* nasal permeation. Histopathological analysis exhibited significantly superior drug targeting, introducing these transferosomes as possible strategy for DZP N2B delivery [90].

Ethosomes are also vesicular NPs made up of about 10% ethanol, which can enhance the solubility of poorly soluble

drugs. Huperzine-A ethosomal gel and cream were developed to benefit from its memory-enhancing ability for possible transdermal AD therapy. Ethosomes were developed by injection method and achieved acceptable drug encapsulation. Results showed better permeation and skin retention than ordinary gel or cream. Findings could support the use of ethosomes for transdermal Huperzine-A delivery [91].

From the vesicular systems, novasomes are vesicular carriers formed of phospholipids, ethanol, together with penetration enhancers. Linagliptin (LGP) novasomes were prepared for IN brain delivery for combating AD due to neuroprotective effect of LGP. Novasomes were prepared by thin film hydration. Cholesterol, stearic acid as well as Span-80 levels were optimized using Box-Behnken design. *Ex-vivo* permeation confirmed higher permeation across camel nasal mucosa compared to LGP solution [92].

Astaxanthin's (AST) therapeutic potential in AD was enhanced by the use of invasomal carriers (liposomes incorporating terpenes). AST-loaded invasomes were developed using cineole as a penetration enhancer in order to obtain stable vesicles with high EE and good stability. Dietary supplementation of AST invasomes enhanced learning and memory in AD animal model. AChE was effectively inhibited in animals' brains. Moreover, brain recovery proteins were successfully restored by the developed invasomes. These modulations strengthen the use of AST invasomes for promising outcomes in AD treatment [93].

Bilosomes or bile-salt-based vesicular carriers, are recently recognized as formulation approach to improve *in vivo* efficacy of various drugs. Bilosomes have considerable stability as compared to liposomes and niosomes. IN delivery system based on luteolin bilosomes was developed to combat AD. Different bilosomes were prepared using various cholesterol-to-phospholipid ratios and different bile salt concentrations. Results confirmed accepted particle size, luteolin encapsulation and suitable stability. *In vivo* experiments on AD mouse model confirmed improved memory after bilosomal administration. Luteolin bilosomes also inhibited amyloid β aggregation and decreased hyperphosphorylated tau [94].

Cubosomes were also designed for brain targeting of piperine after oral use of piperine Tween 80-decorated cubosomes to combat AD. *In vivo* study was performed on AD rat model (sporadic dementia). *In vivo* studies confirmed enhanced cognitive functions and reduced AChE activity for the NPs highlighting their potential to hinder AD progression [95].

Exosomes are nanosized vesicular systems originating from different cell types and are prominent in most biological fluids. Their decreased size and intrinsic ability to facilitate cellular communication introduce exosomes as drug delivery vehicles that deliver drugs to a variety of organs, including brain. Exosomes can deliver bioactive substances

across BBB and hence can be adopted as natural carriers for AD medicaments. Qi *et al.* developed plasma-derived exosomes for enhanced quercetin brain delivery. Quercetin efficacy was also supported by inhibiting Tau phosphorylation in AD mice. Moreover, NFTs were reduced as compared with free quercetin [96].

Nanoemulsions are colloidal systems with higher solubilization capacity. Nanoemulsions were also investigated for AD therapeutic delivery. AChE inhibition activity of rosemary oil was utilized to provide additional benefits to donepezil via their combining in nanoemulsion for subsequent IN administration. Nanoemulsion was first prepared by low-energy emulsification using surfactant (Tween 80) and co-surfactant (diethylene-glycol-monoethyl ether). *In vitro* permeation studies showed enhanced drug penetration from nanoemulsion as compared to plain DZP. Findings suggested that this formulation can be used for futuristic therapy for AD [97]. Also, quercetin nanoemulsion was developed and evaluated for AD treatment after intraperitoneal injection to AD rat model. Results showed enhanced antioxidant as well as anti-inflammatory efficacy in group receiving quercetin nanoemulsion making this formula a promising carrier for AD treatment [98].

Amylolipid nanovesicles, new starch-modified complex NPs, were developed as nanocarriers for curcumin CNS targeting via N2B delivery to tackle AD. Curcumin could inhibit A β aggregation and inflammation. Curcumin was effectively encapsulated by amylolipid vesicles. Brain concentrations of curcumin after IN intranasal administration were markedly higher than those obtained by curcumin IV administration [99].

Organic Polymeric NPs

Different natural and synthetic polymers were used for the preparation of NPs for targeted drug delivery. Organic polymeric NPs include polymeric NPs, nanocapsules, nanomicelles, dendrimers and nanogels. Polymeric NPs can be developed using various polymers such as chitosan and polylactic-co-glycolic acid (PLGA). For efficient rivastigmine encapsulation, NPs were developed by modified nanoprecipitation technique using PLGA as a polymer and polyvinyl alcohol (PVA) as stabilizer in a study by Sánchez-López and colleagues. The developed RIV NPs were evaluated for possible efficacy against AD. *In vivo* evaluation in animal model revealed that RIV NPs could be effective against AD. Animals treated with NPs exhibited significantly ($p < 0.01$) lower brain AChE activity compared to untreated animals. Hence, these PLGA NPs could be ideal carriers for drugs treating AD [100]. Memantine PEG-PLGA NPs were also developed for MEM brain targeting after oral administration. The biocompatibility of these NPs was confirmed by cell lines. The developed NPs had BBB penetration ability as indicated by *in vitro*

and *in vivo* studies. Behavioral tests demonstrated enhanced efficacy compared to MEM solution. Moreover, histopathology studies confirmed diminished β -amyloid plaques as well as decreased neuroinflammation [101].

Donepezil PLGA-NPs were also prepared using double emulsion solvent evaporation, where varying concentrations of PVA and different sonication times were varied to optimize NP characteristics. NPs had acceptable encapsulation efficiency along with sustained release profile. Anti-AChE activity was confirmed in biological activity studies [102].

Alendronate-loaded chitosan nanoparticles were prepared for brain targeting after IN dosing in order to reduce peripheral side effects associated with alendronate. These NPs acquired better pharmacokinetics than IN drug solution. In AD animal model, intranasal NPs markedly altered neurobehavioral and neurochemical changes in mice more than IN drug solution. Brain-targeted delivery of these nanoparticles could be suitable therapeutic option for AD [103].

Nanocapsules are another form of NPs that are formed of lipid core (liquid or solid) within a polymeric shell. With the aim to benefit from curcumin in AD, it was successfully encapsulated in the form of nanocapsules (NCs) using polycaprolactone, triglyceride oil (caprylic/capric) as well as sorbitan monostearate. In AD animal model (induced by streptozotocin), oral NCs markedly reversed memory deficits and attenuated elevated AChE activity caused by STZ. Findings suggested the use of cur NCs to improve memory in AD rat model [104].

Nanofibers (NFs) were also utilized for peptide delivery in a study by Tanaka and coworkers, where brain-permeable peptide NFs were developed for AD. The developed NFs were able to suppress neurotoxicity induced by amyloid β . Moreover, the IV or IN-administered NFs improved cognitive function in AD mouse model, highlighting the potential efficiency of these NFs for treatment of AD [105].

Dendrimers are hyper-branched structures formed of initiator core from which repetitive units are bonded in a tree-like architecture. Dendrimers can also be used for brain-targeted memantine delivery for use in AD mice. IV administration of developed polyamidoamine (PAMAM) lactoferrin-conjugated dendrimers containing MEM achieved enhanced biodistribution in rat model as illustrated by brain uptake of NPs compared to free drug. *In vivo* study in AD mice (AlCl_3 -induced) showed significant improvement in terms of behavioral responses suggesting improved brain bioavailability for MEM [106].

Another form of polymeric NPs is nanogel, which is crosslinked hydrogel particles in submicron scale. In an attempt to enhance the activity of tannic acid together with L-lysine against AD, nanogel formulations were developed and studied. The obtained particles were in the nanometer range, and AChE inhibition activity was obtained at concentration-dependent manner [107].

Hybrid (lipid-polymer) NPs were also used to target rivastigmine to brain. NPs were prepared by film hydration method using varying lipid-to-polymer ratios, and the selected formula was administered to animals. Results confirmed enhanced cognitive functions in animals receiving formulation, suggesting neuroprotective activity [108].

Carbon-based Nanoparticles

Carbon-based NPs include carbon dots and carbon nanotubes. Carbon dots (CDs) attracted considerable attention owing to their nontoxicity, high drug-loading ability and ability to pass BBB. For the delivery of metformin and citric acid as possible therapy for AD, carbon dots (CDs) were prepared. Results showed that intraperitoneal injection of these CDs to AD mice models enhanced neurogenesis, inhibited glial cell activation and reduced $\text{A}\beta$ deposition. Hence, these CNs emerge as promising AD therapy candidates [109]. Carbon nanotubes (CNTs), which are fullerene assemblies with high drug loading potential were used for berberine delivery with polysorbate as well as phospholipid coating to be used for AD. *In vivo* performed pharmacokinetic investigation in rats (IV administration) showed significantly improved bioavailability compared to free berberine. Behavioral assessment (Morris Maze test) showed enhanced effectiveness of the formed CNTs. Moreover, coated CNTs showed marked recovery in memory performance in animals [110].

Surfactant-based Systems

Surfactant-based systems are delivery systems in which surfactants are self-aggregated, almost in the presence of water, to form structures of various properties. Spanlastics are identified as biodegradable surfactant-based nanocarriers that are more stable than liposomes. Based on the promising role of epigallocatechin gallate in AD, EGCG spanlastics for oral use were developed by ethanol injection, where different factors (surfactant type and ratio and surfactant-to-edge activator ratio) were investigated to optimize entrapment efficiency and particle size (PS). Pharmacokinetic evaluation showed improved bioavailability of EGCG compared to either conventional niosomes or free EGCG [111].

Protein-based Nanoparticles

Protein NPs are recently recognized for the delivery of pharmaceuticals, including AD therapeutics, owing to their unique biodegradability and low toxicity [112]. Albumin-chitosan NPs were prepared for curcumin delivery

to modulate curcumin permeability across BBB. Results showed that these NPs had higher permeation across BBB compared to curcumin. Moreover, the developed NPs enhanced A β phagocytosis [113]. In another study, albumin was used as nanocarrier for N2B R-flurbiprofen delivery for AD. R-flurbiprofen has the ability to alleviate mitochondrial dysfunction. Findings demonstrated elevated brain-to-plasma ratio compared to either IN or oral flurbiprofen solution [114].

Metallic Nanoparticles

Metal nanoparticles are explored extensively in biomedical applications due to their unique small size-to-volume ratio as well as their high thermal stability. Gold NPs are promising opportunities for neurodegenerative disorders in which drug passage across BBB is essential. Gold NPs found many applications in treatment and diagnosis of neurodegenerative diseases. In AD, gold NPs work by inhibiting amyloid fibrils, which are an important marker in AD tissues [115]. Efficacy of gold NPs was studied in AD animal model (streptozotocin-induced). Results showed that gold NPs were able to reverse mitochondrial dysfunction, neuroinflammation, as well as oxidative stress in experimental animals. Moreover, gold NPs prevented memory defects in AD animal models. Results highlight the consideration of gold NPs for AD therapy [116].

In another interesting work, selenium-chondroitin sulfate NPs were developed and their therapeutic efficacy was studied. NPs effectively reduced A β aggregation *in vitro*, thus providing neuroprotective effect. Moreover, these NPs decreased reactive oxygen species (ROS) levels and attenuated tau hyperphosphorylation *in vitro* by regulating GSK-3 β expression. So, selenium NPs could be potent multi-functional therapeutic agents for AD [117].

Cerium oxide (antioxidant) NPs have also recently emerged as candidates for AD therapy via IN administration. *In vivo* behavioral investigations in AD rat model (scopolamine-induced) confirmed the ability of these NPs to reverse cognitive ability compared to group treated by RIV patch. Hence, N2B delivery of these NPs could be prospective therapy for AD [118].

AD Treatment Development; Regulatory Issues & Clinical trial Challenges

Despite the great advances in the development and utilization of therapeutics, drug delivery carriers and novel dosage forms, limited therapeutics are being approved for AD management. This is due to the fact that, in addition to exploration of drug safety and efficacy, important issues need to be addressed by regulatory agencies to confirm that patients can receive appropriate medicines

[119]. Regulatory agencies, including European Medicines Agency (EMA), FDA and others, have significant role in advancement of new therapeutic drug development. The collaboration between regulatory agencies and pharmaceutical industries is important to facilitate innovative drug development and ensure availability of new therapeutics for patients [120].

In addition to absence of limited evidence for particular target for therapeutic intervention, the lack of well-defined regulations that govern utilization of newly developed formulations contributes to the complexity of AD novel treatment approval. While the use of nanomedicine can solve many problems encountered in AD therapy, lack of regulations limited the clinical potential of nanomedicines. So, engagement of different regulatory bodies is needed to establish robust foundation for nanocarrier development [121]. Regulatory agencies demand detailed NP characterization, including investigation of their physicochemical properties and pharmacokinetics, as well as interactions with biological systems. In addition, regulatory agencies require thorough toxicity investigations of nanoparticles, which are costly and time-consuming which in turn, slows down development process [48].

In addition to the regulatory limitations, the development of disease modifying therapy for AD can also be hindered by preclinical evaluation difficulties. Animal models and evaluation frameworks often fail in accurately reflecting interactions of multiple targets [122].

Additional challenges for treatment development are those challenges associated with conducting clinical trials. AD clinical trials are known to be slower to include participants, take longer for completion, and are highly expensive compared to trials for other therapeutic areas. These trials can take several years to observe clinical effects and demonstrate delayed AD progression. Moreover, the investigation of drug efficacy in predementia stages of AD is more complex than studying drug effect in advanced disease where, clinical symptoms could be clearly manifested [120]. The lack of biomarkers that accurately detect progression of AD adds more complexity for conducting clinical trials. Conducting clinical trials for AD therapeutics also faces ethical challenges. Clinical trials mandate involvement of vulnerable people, as those with cognitive impairments from whom informed consent and risk–benefit evaluations require ethical considerations. Moreover, AD not only affects AD patients, but also tends to affect the lives of patients' caregivers and families. Hence, specific attention should be given to them while conducting AD clinical trials [1]. For possible injuries, recent guidance by National Alzheimer's Coordinating Center (NACC) recommends participant reimbursement in case of trial-related injuries [51, 123].

Future Perspectives

While AChE inhibitors and NMDA receptor antagonists remain the standard AD therapy, additional emerging technologies are taken in consideration to prevent AD progression and provide successful AD cure.

Nanobubbles

Nanobubbles are currently being explored for drug, gas and gene delivery. The combination of focused ultrasound with nanobubbles has great potential for BBB opening because of its targeting ability and noninvasive nature. This technique was reported to induce neurogenesis and reduce amyloid load in AD. However, biosafety, complex production and the ability to control pharmacokinetics have to be addressed to promote the use of nanobubbles as drug delivery carriers [124].

Immunization

Recent research has been shifted to the prevention of AD via immunization. Preclinical research is conducted on mouse models to investigate the immunization outcomes. Results showed reduction of amyloid-beta and tau pathology. However, cognitive outcomes are still inconsistent [125].

Stem Cell

Stem cell-based therapy presents an alternative approach for AD treatment. Stem cells can divide into specialized adult cells with self-renewal ability and multiplication. Stem cells including embryonic stem cells (ESCs) as well as mesenchymal stem cells (MSCs) were found to have the capacity to reverse ACh levels, decrease phosphorylated tau and neuroinflammation. However, additional research is needed before stem cell therapy can be utilized as clinically feasible AD treatment [126].

Gene Therapy

Gene therapy for AD has gained a lot of interest recently. In AD animal models, brain growth factor can inhibit neuronal death, build up new synapses and potentiate neuronal functions, including learning and memory [127]. In a study by Zhang *et al.*, basic fibroblast growth factor (bFGF) encapsulated NPs were developed and decorated with *Solanum tuberosum* lectin (carbohydrate-binding protein) for brain targeting. IN administration of these NPs achieved adequate brain distribution. Neuroprotective

effect was also confirmed for NPs in Morris water maze test in rats. Moreover, spatial learning and memory were markedly improved in AD rats. Histopathology study confirmed the safety of these NPs [128].

Neural Implant

Neural implants are medical devices to be inserted into brain for treatment or management of various medical conditions. The promising effects of neural implants as drug delivery systems in studies conducted on rodent model led to the suggestions for their use for neurodegenerative disorders affecting elderly such as Alzheimer's disease [129].

Artificial Intelligence in AD

The artificial intelligence (AI) can help in the diagnosis as well as the treatment of AD. Cognitive and behavioral investigations could benefit from AI as a tool to identify and track AD progression. Personalized treatment could also be revolutionized by AI-driven approaches. Moreover, drug screening and repurposing can be guided by implementation of suitable predictive modeling. In conclusion, AI-driven advancements can transform AD therapy from an incurable progressive disease to more manageable condition, thus improving healthcare options. However, data privacy, ethical considerations, and the need for AI integration tools remain challenges for AI application in AD [130, 131].

Conclusion

The dramatic rise in AD prevalence and the inadequacy of current treatments necessitate the development and implementation of new therapies. While recent researches provided more detailed understanding of AD mechanisms, appropriate treatment is still challenge. To date, all approved medicaments can also deliver symptomatic relief. Hence, new medicaments and drug delivery systems that stop or reverse AD progression and simplify administration are essential.

Several treatments are encountered in clinical trials. Specific attention has to be given to the dosage form and the route of administration. In this concern, the importance of intranasal delivery as non-invasive method for drug delivery to the brain was highlighted. Moreover, the potential of nanomedicine has solved problems encountered in drug targeting to CNS via facilitating drug passage across BBB. Nanomedicine can hence be tailored for therapeutic delivery for AD with great efforts required for translating these NPs

from lab to the market. In this concern, regulatory agencies require comprehensive toxicity evaluations that could be costly and time-consuming, leading to slowing of the development process.

Recent studies have introduced immunotherapeutic, gene therapy and stem cell therapy as disease-modifying therapies with promising results regarding the ability to restore neuronal functions. However, continual research in the use of novel drug delivery with suitable delivery carriers, along with addressing biocompatibility of such delivery techniques, would increase the hope for better AD management.

Abbreviations AD: Alzheimer's disease; A β : β -Amyloid; NFTs: Neurofibrillary tangles; BBB: Blood brain barrier; EOD: Early-onset dementia; APOE: Apolipoprotein E; NMDA: N-methyl-D-aspartate; AChE: Acetylcholine esterase; APP: Amyloid precursor protein; ROS: Reactive oxygen species; CNS: Central nervous system; DPZ: Donepezil; RIV: Rivastigmine; GAL: Galantamine; MEM: Memantine; GSK: Glycogen Synthase Kinase; BACE: β -Site amyloid cleaving enzyme; P-gp: P-glycoprotein; ER: Extended-release; ODTs: Orally disintegrating tablets; ODFs: Oral disintegrating films; MDFs: Mouth dissolving films; MNs: Microneedles; IN: Intranasal; N2B: Nose to brain; NPs: Nanoparticles; SLNs: Solid lipid nanoparticles; NLCs: Nanostructured lipid carriers; Tf: Transferrin; CPP: Cell-penetrating peptide; LP: Ligustrazine phosphate; LGP: Linagliptin; AST: Astaxanthin; PLGA: Poly-lactic-co-glycolic acid; PVA: Polyvinyl alcohol; NCs: Nanocapsules; PAMAM: Polyamidoamine; CDs: Carbon dots; CNTs: Carbon nanotubes; EGCG: Epigallocatechin gallate; LLC: Lyotropic liquid crystalline; ESCs: Embryonic stem cells; MSC: mesenchymal stem cells; bFGF: basic fibroblast growth factor; AI: Artificial intelligence

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Declarations

Conflict of interest Author declares no conflicts of interest.

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